COMMUNITY SERVICE PROJECT REPORT ON
WATER SEGRETION & RAINWATER HARVESTING

Submitted in partial fulfillment of requirements for the award of degree of

BACHELOR OF TECHNOLOGY IN COMPUTER SCIENCE & ENGINEERING

Submitted By

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Under the Esteemed Guidance of

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Assistant Professor



NARASARAOPETA
ENGINEERING COLLEGE

DEPARTMENT OF COMPUTER SCIENCE & ENGINEERING

NARASARAOPETA ENGINEERING COLLEGE: NARASARAOPET

(AUTONOMOUS)

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Yallamanda Village,NARASARAOPET -522601

2026-2027

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(AUTONOMOUS)

DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING

BONAFIDE CERTIFICATE

This is to certify that the CSP project work entitled "WATER SEGREGATION&RAIN WATER HARVESTING" is a Bonafide work done by "Tadepalli Uma Naga Vyshnavi (24471A05CV)" in partial fulfillment of requirement for the award of the degree of Bachelor of technology in the department of COMPUTER SCIENCE AND ENGINEERING of the NARASARAOPETA ENGINEERINGCOLLEGE, NARASARAOPET, during the year 2026-2027.

Project Guide

Y.Chandana, M.Tech

Assitant Professor

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Dr.S.N.Tirumala Rao,M.Tech ,Ph.D

Professor & HOD

EXTERNAL EXAMINER

Program Book

CommunityServiceProject

APSTATECOUNCILOF HIGHER EDUCATION

(A STATUTORY BODY OF GOVERNMENT OFANDHRA PRADESH)

COMMUNITY SERVICE PROJECT

STUDENT AND PROJECT DETAILS

Name of Student	Tadepalli Uma Naga Vyshnavi
Name of College	Narasaraopeta Engineering College
Registration Number	24471A05CV
Period of CSP	MAY 2026 TO JULY 2026
Project Title	WATER SEGREGATION & RAINWATER HARVESTING
Name of Community / Habitation	Kondakavuru
Address of Habitation	Kondakavuru, Palnadu District

PROJECT LOCATION

The Community Service Project was carried out in Kondakavuru, Palnadu District, during the period stated above.

The details above are arranged in a clear, consistent format for the project record.

WATER SEGRETION&RAINWATER HARVESTING

Submitted in accordance with the repuirement for the degree of B.TECH-CSE

Nameof College	: Narasaraopeta Engineering Collage
Department	: Computer Science & Engineering
Name of the Faculty	: Y.Chandana
Duration of the CSP	: MAY2026 TO JULY2026
Student	:Tadepalli Uma Naga Vyshnavi
Program Study	: Computer Science & Engineering
Year of Study	: III B.TECH
Roll no	: 24471AOSCV
Date of the Submission	:

Student's Declaration

I’m Tadepalli Uma Naga Vyshnavi a student of B.Tech. Program, Reg. No. 24471AO5CV of the Department of Computer Science and Engineering, Narasaraopeta Engineering College do hereby declare that I have completed the mandatory Community Service Project on ***'WATER SEGRETION & RAIN WATER HARVESTING"*** from May-2026tojuly-2026

Faculty Guideship of Mrs. Chandana, Department of Computer Science and Engineering in Narasaraopeta Engineering College.

Signature of the Student and Date

Faculty Guide
(Y.Chandana M.Tech)

Head of Department.
(Dr.S.N. Tirumala Rao , M.Tech, Ph.d)

Certificate from Official of the Community

This is to certify that TADEPALLI UMA NAGA VYSHNAVI Reg. No: 24471AO5CV

from the Department of Computer Science and Engineering, Narasaraopeta

Engineering College underwent Community Service in" WATER

SEGRETION&RAINWATER HARVESTING PORTAL" from

The overall performance of the Community Service Project Volunteer during his

Community Service Project is found to be .....

Authorized Signatory with
date and seal

Acknowledgements

We wish to express my thanks to various personalities who are responsible for the completion of this project. We are extremely thankful to our beloved chairman Sri **M. V. Koteswara Rao,** B.Sc., who took keen interest in us in every effort throughout this course. We owe our sincere gratitude to our beloved principal **Dr.S.Venkateswarlu,** Ph.D for showing his kind attention and valuable guidance throughout the course.

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We affectionately acknowledge the encouragement recived from our friends and those who involved in giving valuable suggestions had clarifying our doubts which had really helped us in successfully completing our Community Service Project

Project associates

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INTRODUCTION

Rain water harvesting and waste segregation are two practical community-level measures that support water security, public health, environmental protection, and responsible resource use. Rainwater harvesting captures rainfall before it is lost as surface runoff and directs it toward storage or groundwater recharge. Waste segregation separates materials at the point of generation so that wet waste can be composted, dry waste can be recovered for recycling, and hazardous or sanitary waste can be handled safely.

The Community Service Project connects these technical ideas with local participation. The project team is expected to understand how rainwater flows through the selected community, how water is used and wasted, where recharge or storage could be feasible, and how household waste is currently collected, mixed, dumped, burned, or recovered. The project also emphasizes that infrastructure alone is not sufficient: systems must be maintained, waste must be separated consistently, and community members must understand their role.

This report is designed as a complete academic project framework. It includes detailed background, methods, activity planning, weekly documentation, observations, awareness activities, outcomes, and recommendations. Statements that depend on field verification are marked with square brackets and must be replaced with actual records.

OBJECTIVES OF THE PROJECT

- *To understand the principles and local relevance of rooftop rainwater harvesting, surface runoff collection, storage, filtration, and groundwater recharge.*

- *To identify existing water-use patterns, drainage pathways, possible harvesting locations, and maintenance requirements in the selected community.*

- *To study household and community waste generation, collection, disposal, and recovery practices.*

- *To promote segregation of wet, dry, recyclable, sanitary, and hazardous waste at the point of generation.*

- *To explain composting, recycling, responsible disposal, and the relationship between unmanaged waste and water pollution.*

- *To conduct community interaction and awareness activities using simple, respectful, and locally relevant communication.*

- *To develop a realistic action plan that can be monitored by residents, local institutions, and community volunteers.*

➢ *The main objectives of rainwater harvesting are:*

1. ***To conserve water*** *– Save rainwater for future use*

.

2. ***To recharge groundwater*** *– Increase the groundwater level by allowing rainwater to seep into the soil.*

3. ***To reduce water scarcity*** *– Provide water during dry seasons and droughts.*

4. ***To prevent soil erosion*** *– Reduce the flow of rainwater over the land.*

5. ***To reduce flooding*** *– Minimize surface runoff and waterlogging*

6. ***To improve groundwater quality*** *– Help maintain groundwater levels and reduce the effects of seawater intrusion in coastal areas.*

7. ***To provide water for domestic use*** *– Use collected rainwater for washing, gardening, and other suitable purposes.*

8. ***To support agriculture*** *– Provide water for irrigation and livestock.*

9. ***To reduce dependence on municipal water supplies*** *– Supplement the regular water supply.*

10. ***To protect the environment*** *– Promote sustainable use of natural water resources.*

➢ IMPORTANCE OF RAINWATER HARVESTING

Rainfall is a seasonal resource. In many settlements, a large part of rainwater flows quickly across roofs, roads, and open land into drains or low-lying areas. A rainwater harvesting system slows, filters, stores, or infiltrates this flow. It can reduce dependence on external water supplies, supplement non-potable uses, support groundwater recharge, reduce local waterlogging, and encourage conservation-oriented behaviour.

- Water security: stored rainwater can supplement cleaning, gardening, toilet flushing, or other non-potable uses where appropriate.
- Groundwater support: recharge structures can help rainwater infiltrate into suitable subsurface formations after adequate filtration and site assessment.
- Reduced runoff: directing roof and surface runoff safely can reduce erosion, ponding, and pressure on stormwater drains.
- Economic value: using harvested rainwater for suitable purposes can reduce demand on purchased or pumped water, subject to local feasibility.
- Community resilience: decentralised systems provide a practical response to irregular rainfall and seasonal water stress.

Environmental awareness: visible systems make water conservation a daily community practice rather than an abstract message.

MAINLY :

1. **Water conservation:** Saves rainwater for future use.
2. **Groundwater recharge:** Increases the groundwater level.
3. **Prevents water scarcity:** Provides water during dry seasons and droughts.
4. **Reduces flooding:** Decreases surface runoff and waterlogging.
5. **Prevents soil erosion:** Reduces the washing away of fertile soil.
6. **Supports agriculture:** Provides water for irrigation and gardening.
7. **Reduces water bills:** Decreases dependence on municipal water supplies.
8. **Eco-friendly:** Helps protect natural water resources.
9. **Improves water availability:** Provides an additional source of water for domestic purposes.
10. **Sustainable development:** Helps ensure water availability for future generations.

Rainwater harvesting is an effective method of conserving water, Recharging ground water,and maintaining ecological balance

TYPES AND METHODS OF RAINWATER HARVESTING

1. Rooftop rainwater harvesting:

Rain falling on roofs is collected through gutters and downpipes. A first-flush arrangement diverts the initial flow carrying dust and roof debris. The remaining water may pass through a filter and be directed to a storage tank or recharge structure.

2. Surface runoff harvesting:

Runoff from paved areas, courtyards, roads, open grounds, and slopes can be guided through channels, silt traps, filtration beds, recharge pits, trenches, percolation structures, or ponds. The design must consider slope, soil, contamination risk, and safe overflow.

3. Storage-based harvesting:

Rainwater is stored in tanks, drums, sumps, underground reservoirs, or community-level storage structures. Tanks should be covered, protected from contamination, and cleaned at planned intervals.

4. Groundwater recharge:

Recharge pits, recharge trenches, soakaways, percolation pits, recharge wells, and infiltration structures allow filtered water to enter the ground. Recharge should not be used blindly where water quality risks, contaminated sites, or unsuitable geology are present.

5. Farm and landscape measures:

Contour bunds, vegetated swales, farm ponds, check dams, and graded channels can slow runoff and improve infiltration in suitable rural or peri-urban areas.

COMPONENTS OF A RAINWATER HARVESTING SYSTEM

Component	Function	Verification / Field Note
Catchment surface	Roof or surface that receives rainfall	Roof type
Gutters and downpipes	Collect and convey roof runoff	Existing
Leaf screen / mesh	Stops leaves and larger debris	GOOD
First-flush diverter	Diverts initial dirty runoff	Present
Filter unit	Removes suspended matter before storage or recharge	FILTER PAPER
Storage tank	Stores water for later use	20,000SQ METERS
Recharge structure	Directs filtered water into the ground	FEASIBLE
Overflow arrangement	Prevents flooding when storage is full	ARRANGEMENT OF PIPES
Tap / outlet	Allows controlled use of stored water	OUTFLOW
Maintenance schedule	Keeps the system functional and hygienic	WORKER

Importance of Water Segregation

1. Protects the Environment

Waste segregation helps reduce environmental pollution. When waste is separated properly, less garbage is
dumped in open areas and landfills. It also helps prevent the contamination of soil, water, and air.

2. Improves Recycling

Waste segregation makes recycling easier and more effective. Materials such as paper, plastic, glass, and
metals can be separated and sent for recycling. These materials can be used to manufacture new products,
reducing the need for raw materials.

3. Reduces Landfill Waste

When biodegradable and recyclable materials are separated, the amount of waste sent to landfills decreases.
This saves land, reduces unpleasant smells, and helps prevent pollution caused by improperly managed waste.

4. Promotes Public Health

Mixing all types of waste can attract flies, mosquitoes, rats, and other disease-carrying organisms. These
organisms can spread diseases such as malaria, dengue, and other infections. Proper segregation and
disposal of waste help maintain hygiene and protect public health.

5. Helps in Composting

Wet waste, such as vegetable peels, fruit scraps, leftover food, and leaves, can be separated and converted
into compost. Compost is a natural fertilizer that improves soil fertility and supports plant growth. This also
reduces the amount of organic waste sent to landfills.

6. Conserves Natural Resources

Recycling paper, plastic, glass, and metals reduces the need to extract new raw materials from nature. For
example, recycling paper helps reduce the demand for wood, while recycling metals reduces the need for
mining. Thus, waste segregation helps conserve forests, minerals, and other natural resources.

7. Saves Energy

Manufacturing products from recycled materials often requires less energy than producing them from raw materials. Proper segregation helps recover recyclable materials and reduces energy consumption in industries.

8. Reduces Greenhouse Gas Emissions

Organic waste that decomposes in landfills can produce methane, a powerful greenhouse gas. Separating
organic waste for composting or other suitable treatment can help reduce these emissions and support
climate protection.

9. Ensures Safe Disposal of Hazardous Waste

Some waste materials, such as batteries, electronic waste, medicines, chemicals, and certain cleaning products,
may contain harmful substances. Separating them from ordinary household waste helps prevent the release
of toxic materials into the environment and supports safe handling and disposal.

10. Keeps Surroundings Clean

Waste segregation helps maintain cleanliness in homes, schools, colleges, streets, parks, and other public
places. It reduces littering, unpleasant smells, and the accumulation of garbage. Clean surroundings also
create a healthier and more pleasant living environment.

11. Reduces Waste Management Costs

When waste is properly separated, recyclable materials can be recovered more easily, and less waste needs to
be transported to landfills. This can reduce the cost of waste collection, transportation, and disposal.

12. Creates Employment Opportunities

Waste segregation supports employment in recycling centres, composting units, waste collection services,
and waste treatment facilities. It also encourages the development of industries that use recycled materials.

13. Encourages Responsible Behaviour

Waste segregation develops good habits among people. It teaches individuals to take responsibility for the waste they generate and encourages the principles of **Reduce, Reuse, and Recycle (3Rs)**. These habits are
especially important for children and students.

14. Supports Sustainable Development

Sustainable development means meeting present needs without compromising the ability of future generations to meet their own needs. Waste segregation reduces resource wastage, limits pollution, and promotes the efficient use of materials. Therefore, it plays an important role in building a sustainable society.

Methods of Water Segregation

Water segregation is the process of separating different types of water based on their source, quality, or use. It helps prevent pollution, conserve freshwater, and promote the reuse of water.

The main methods of water segregation are:

1. Rainwater Separation

Rainwater is collected separately from rooftops and other surfaces using gutters and pipes. It can be stored in tanks or directed into recharge pits for groundwater replenishment.

2. Grey water Separation

Wastewater from bathrooms, washbasins, and laundry is collected separately from toilet wastewater. After suitable treatment, it can be reused for gardening, flushing toilets, and other non-drinking purposes.

3. Black water Separation

Wastewater from toilets is collected separately because it contains human waste and harmful microorganisms. It is sent for appropriate treatment before safe disposal or reuse.

4. Industrial Waste water Separation

Industrial wastewater is collected separately from domestic wastewater because it may contain chemicals, oils, heavy metals, and other pollutants. It requires suitable treatment before discharge or reuse.

5. Storm water Separation

Rainwater flowing over roads, rooftops, and open spaces is collected through separate drainage systems. This helps reduce flooding and prevents stormwater from mixing unnecessarily with sewage.

6. Oil and Water Separation

Oil and water mixtures from garages, workshops, and industries are separated using oil separators. This prevents oil from entering drains, rivers, and lakes.

7. Clean Water and Waste water Separation

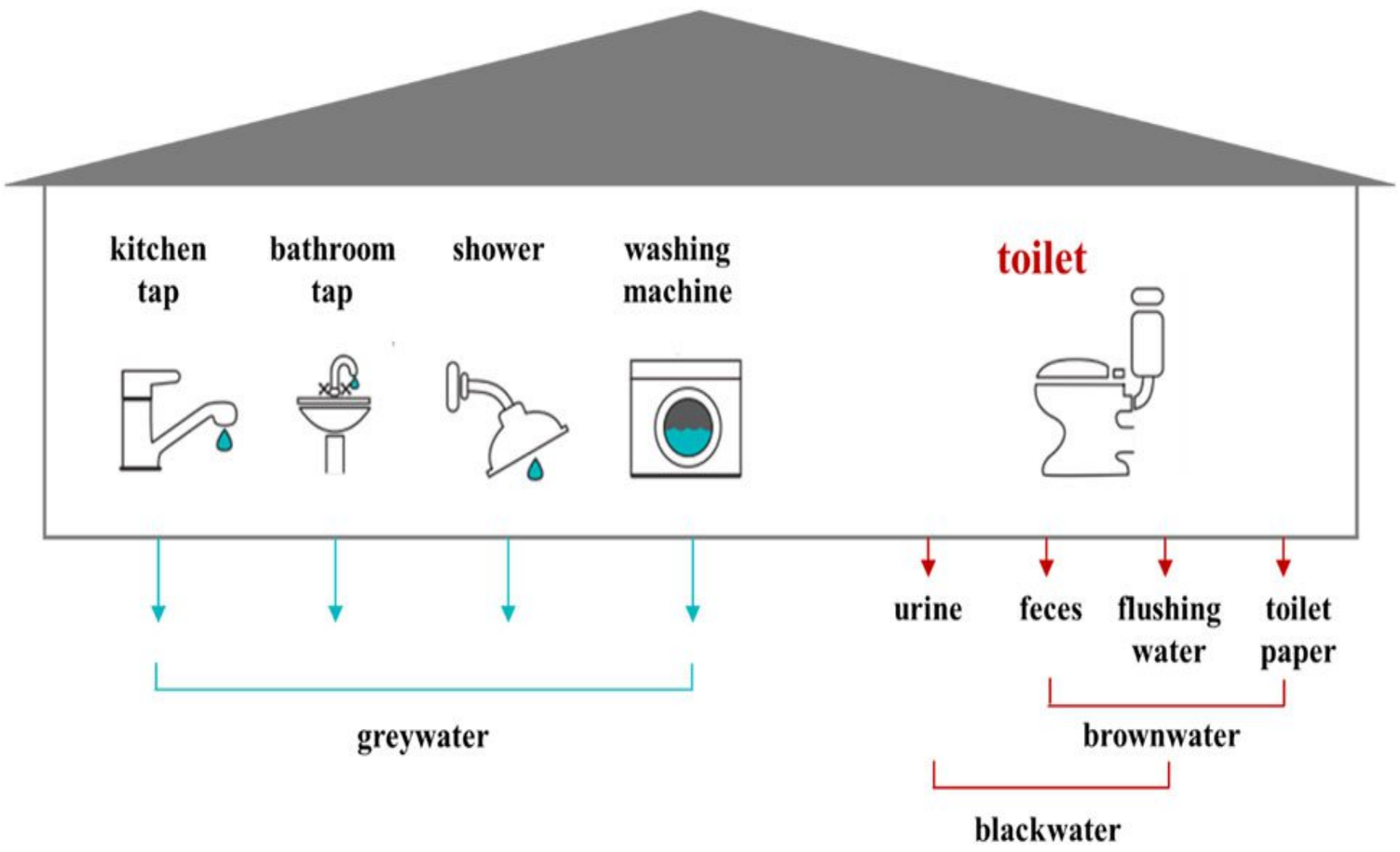
Pipes carrying clean water are kept separate from wastewater drainage pipes. This prevents contamination of drinking water and protects public health.

. Water Separation by Treatment

Water containing different impurities can be treated using methods such as filtration, sedimentation, and other suitable treatment processes. These methods help separate contaminants and improve water quality.

Waste category	Typical examples	Preferred handling
Wet / biodegradable	Food scraps, vegetable peels, tea waste, garden leaves	Home/community composting or authorised wet-waste processing
Dry recyclable	Paper, cardboard, clean plastic, metal cans, glass	Keep dry and send to authorised collector / material recovery facility
Dry non-recyclable	Contaminated multilayer packaging or materials without a recovery route	Follow local-authority collection and disposal instructions
Sanitary waste	Diapers, sanitary pads, bandages, contaminated tissues	Wrap, label where required, and hand over through the authorised system
Domestic hazardous waste	Batteries, bulbs, paint containers, chemicals, e-waste components	Store separately and use authorised collection / take-back channels
Construction / bulky waste	Rubble, broken furniture, large household items	Do not place in regular bins; use designated collection arrangements


Grey Water Sources
Yellow Water Source
Mixture of Grey
&
Yellow water
Black Water


kitchen tap
bathroom tap
shower
washing machine
toilet
urine
feces
flushing water
toilet paper
greywater
brownwater
blackwater

Project Planning and Methodology of Water Segregation

1. Project Planning

Project planning is the process of deciding the objectives, activities, materials, and steps required to
complete a project successfully.

The main aim of this project is to understand the importance of water segregation and develop methods for separating different types of water for effective management and
reuse.

Objectives of the Project

To understand the importance of water segregation.

To identify different sources and types of water.

To separate rainwater, greywater, blackwater, and industrial wastewater.

To prevent the mixing of clean water with contaminated water.

To promote the reuse and recycling of wastewater.

To reduce water pollution and conserve freshwater resources.

To create awareness about sustainable water management.

Activities Involved in Project Planning

- **Problem identification:** Identify problems caused by mixing clean water and wastewater.
- **Data collection:** Collect information from books, websites, and observations.
- **Resource identification:** Arrange materials such as pipes, containers, labels, and charts.
- **System design:** Prepare a layout showing how different types of water can be collected separately.
- **Implementation:** Demonstrate a simple water segregation system.

Evaluation: Observe the system and identify its benefits and limitations

WEEK 1: COMMUNITY INTRODUCTION AND PROBLEM IDENTIFICATION

Objective of the Activity Done:

To introduce the project, understand the community context, and identify initial concerns related to water

conservation, rainwater flow, waste disposal, and segregation.

Detailed Work Description:

The team will meet community representatives and explain the purpose, duration, and limits of the

project. Initial discussions will focus on water sources, seasonal shortages, roof drainage, stagnant water,

waste collection, open dumping, burning, and the presence or absence of separate containers. The team

will conduct a walk-through without disturbing private property and will note locations requiring later

verification.

Observations and Findings:

The first week establishes trust and prevents the project from making assumptions. The team should

record the names of community contacts, permissions obtained, locations visited, and concerns raised.

Actual attendance and survey counts must be inserted from the field register.

Outcome and Learning:

The team learns how to introduce a technical topic in simple language, ask non-leading questions, respect

community knowledge, and convert broad concerns into projectable issues.

WEEK 1: COMMUNITY INTRODUCTION AND PROBLEM IDENTIFICATION

DAILY ACTIVITY RECORD

DAY & DATE	BRIEF DESCRIPTION OF THE DAILY ACTIVITY	LEARNING OUTCOME	PERSON IN CHARGE / SIGNATURE
Monday 11/05/2026	Conducted a survey on household water use and waste. Asked residents how they dispose of wastewater and manage rainwater.	Identified common water-use habits in the community and learned about residents' awareness of water conservation.	
Tuesday 12/05/2026	Visited nearby homes to observe water storage and drainage. Asked families about water shortages during summer.	Collected information about water availability and storage and understood the need for better water management.	
Wednesday 13/05/2026	Discussed how to separate clean, grey and contaminated water. Recorded common problems in household water use.	Learned the basic idea of water segregation and identified areas where safer water handling can be improved.	
Thursday 14/05/2026	Observed rooftops and open spaces suitable for rainwater collection. Noted places where runoff flows away unused.	Recognized possible rainwater collection areas and understood how harvesting can save water.	
Friday 15/05/2026	Discussed water wastage with community members. Asked about water-related concerns and conservation habits.	Improved understanding of local water issues and identified key topics for awareness activities.	
Saturday 16/05/2026	Summarized survey responses and observations. Prepared simple points for the next awareness activity.	Organized the collected information and set priorities for community awareness about water conservation.	

WEEK 2: WATER USAGE, DRAINAGE, AND EXISTING HARVESTING FACILITIES
Objective of the Activity Done:

To study water-use patterns, rainfall drainage routes, existing storage or recharge facilities, and basic maintenance conditions.

Detailed Work Description:

The team will observe representative houses or institutions, identify roof areas and downpipe routes, and

trace where rainwater currently flows. Discussions will cover water used for drinking, cooking, washing,

gardening, cleaning, and other purposes, without requesting sensitive personal information. Existing

tanks, pits, filters, open drains, blocked channels, and overflow points will be documented through

sketches and photographs where permission is available.

Observations and Findings:

A simple site inventory will be prepared. The team will identify possible harvesting points and note

constraints such as roof condition, space, contamination risk, access, maintenance responsibility, and safe

overflow. No design or capacity should be finalised without a site-specific technical check.

Outcome and Learning:

Students learn to connect water-use behaviour with infrastructure, distinguish storage from recharge, and

understand why maintenance and water quality are part of system design.

WEEK 2: WATER USAGE, DRAINAGE, AND EXISTING HARVESTING FACILITIES

DAILY ACTIVITY RECORD

DAY & DATE	BRIEF DESCRIPTION OF THE DAILY ACTIVITY	LEARNING OUTCOME	PERSON IN CHARGE / SIGNATURE
Monday 18/05/2026	Met residents to discuss water segregation and rainwater harvesting. Listed the main difficulties they experienced.	Understood the community's main water-related concerns and identified topics that needed simple explanations.	
Tuesday 19/05/2026	Explained why wastewater should not be mixed with drinking water. Discussed safe ways to handle different types of water.	Learned the importance of preventing water contamination and improved awareness of safe water practices.	
Wednesday 20/05/2026	Collected information about greywater from kitchens and bathrooms. Discussed where it can and cannot be reused safely.	Recognized possible uses of suitable greywater and learned that contaminated water requires proper treatment.	
Thursday 21/05/2026	Introduced rooftop rainwater harvesting to residents. Explained how gutters and pipes can direct rainwater to storage.	Understood the basic parts of a harvesting system and learned how rooftop collection can save water.	
Friday 22/05/2026	Discussed water-saving habits with families. Encouraged closing taps and repairing leaks promptly.	Identified practical ways to reduce water wastage and learned how small daily actions conserve water.	
Saturday 23/05/2026	Reviewed information gathered during discussions. Prepared simple points for community awareness materials.	Summarized the community's needs and questions and planned clearer explanations for the next stage.	

WEEK 3: RAINWATER HARVESTING METHODS AND SUITABLE LOCATIONS
Objective of the Activity Done:

To compare rooftop, surface-runoff, storage, and recharge approaches and identify locations suitable for

further consideration.

Detailed Work Description:

The team will study technical options and prepare a simple suitability matrix. Candidate locations may

include a school roof, community building, household cluster, open ground, roadside drain, or garden

area, subject to permission and safety. For each candidate, the team will record catchment type, slope,

available space, likely debris load, access for cleaning, storage or recharge preference, and overflow route.

Observations and Findings:

The team will prepare a preliminary concept sketch showing catchment, gutter, first flush, filter, storage

or recharge unit, and overflow. The sketch is a student-level planning aid and must not replace a qualified

engineering design where construction is proposed.

Outcome and Learning:

Students improve technical drawing, site assessment, teamwork, and the ability to explain why one method may be more appropriate than another.

WEEK 3: RAINWATER HARVESTING METHODS AND SUITABLE LOCATIONS

DAILY ACTIVITY RECORD

DAY & DATE	BRIEF DESCRIPTION OF THE DAILY ACTIVITY	LEARNING OUTCOME	PERSON IN CHARGE / SIGNATURE
Monday 25/05/2026	Reviewed rooftop rainwater harvesting and its basic components with reference to the community setting.	Understood how a roof can act as a catchment and how rainwater can be collected safely.	
Tuesday 26/05/2026	Observed suitable roof areas and noted possible routes for gutters and downpipes.	Learned to identify potential collection points and trace the movement of roof runoff.	
Wednesday 27/05/2026	Discussed surface-runoff collection and locations where runoff could be directed safely.	Compared rooftop collection with surface-runoff harvesting and understood the need for safe drainage.	
Thursday 28/05/2026	Studied the purpose of first-flush arrangements, filters, and silt traps.	Learned why initial dirty runoff and suspended particles should be removed before storage or recharge.	
Friday 29/05/2026	Discussed storage tanks, recharge pits, and the need for suitable site conditions.	Distinguished between storing rainwater and allowing filtered water to recharge groundwater.	
Saturday 30/05/2026	Prepared a simple concept sketch showing collection, filtration, storage or recharge, and overflow.	Improved basic planning and learned that final designs require site-specific technical assessment.	

WEEK 4: HOUSEHOLD WASTE DISPOSAL AND SEGREGATION PRACTICES
Objective of the Activity Done:

To understand how households handle wet, dry, recyclable, sanitary, and hazardous waste and to identify

barriers to segregation.

Detailed Work Description:

The team will observe available bins or storage containers and discuss collection frequency, common

waste types, composting, recycling, open dumping, burning, and drain litter. The discussion should focus

on practices rather than judging individuals. Where a survey is used, the team shall record the

questionnaire, sample size, dates, and consent process.

Observations and Findings:

Findings should distinguish between generation, storage, collection, transport, recovery, and final

disposal. The team will note which materials are already recovered by informal or formal collectors and

which categories lack a clear route. Actual percentages must be calculated only from verified survey

sheets.

Outcome and Learning:

Students learn waste classification, ethical field interaction, data recording, and the importance of

designing recommendations around available collection systems.

WEEK 4: HOUSEHOLD WASTE DISPOSAL AND SEGREGATION PRACTICES

DAILY ACTIVITY RECORD

DAY & DATE	BRIEF DESCRIPTION OF THE DAILY ACTIVITY	LEARNING OUTCOME	PERSON IN CHARGE / SIGNATURE
Monday 01/06/2026	Discussed household waste generation and the common methods used for storing and disposing of waste.	Understood the main stages of household waste handling and disposal.	
Tuesday 02/06/2026	Identified wet and dry waste examples through community discussion and observation.	Learned to distinguish biodegradable waste from recyclable and other dry waste.	
Wednesday 03/06/2026	Discussed sanitary and domestic hazardous waste and the need to keep them separate.	Recognized the importance of safe handling and separate collection of special waste categories.	
Thursday 04/06/2026	Asked about waste collection frequency, recycling, composting, and available collection services.	Understood how local collection arrangements influence household segregation practices.	
Friday 05/06/2026	Noted concerns such as mixed waste, open dumping, burning, and litter near drains.	Identified common barriers to proper segregation and their possible effects on the surroundings.	
Saturday 06/06/2026	Summarized observations and prepared practical suggestions suited to available collection systems.	Improved field-note preparation and learned to make realistic, community-focused recommendations.	

WEEK 5: AWARENESS ABOUT RAINWATER HARVESTING AND WATER CONSERVATION

Objective of the Activity Done:

To communicate the value of rainwater harvesting, safe water use, groundwater recharge, and routine

maintenance.

Detailed Work Description:

The team will conduct an awareness activity using a poster, labelled system sketch, demonstration model,

or short talk. Key messages will explain how rain reaches a roof, why the first runoff may be dirty, how

filters work, why storage tanks should be covered, how recharge requires suitable site conditions, and why

luring the rainy season.

Observations and Findings:

The team will record the date, place, audience, materials used, questions asked, and actions suggested by

participants. The activity should end with a short checklist or pledge that residents can apply, such as

checking downpipes, preventing roof contamination, and avoiding water wastage.

Outcome and Learning:

Students develop public-speaking, visual communication, technical explanation, and facilitation skills.

WEEK 5: AWARENESS ABOUT RAINWATER HARVESTING AND WATER CONSERVATION

DAILY ACTIVITY RECORD

DAY & DATE	BRIEF DESCRIPTION OF THE DAILY ACTIVITY	LEARNING OUTCOME	PERSON IN CHARGE / SIGNATURE
Monday 08/06/2026	Introduced the importance of rainwater harvesting and water conservation to community members.	Improved understanding of the value of conserving rainfall and freshwater resources.	
Tuesday 09/06/2026	Explained how rainwater flows from rooftops through gutters and downpipes.	Learned the basic flow path in a rooftop rainwater harvesting system.	
Wednesday 10/06/2026	Discussed first-flush arrangements and the role of filters before water enters storage.	Understood why initial runoff is diverted and why filtration is important.	
Thursday 11/06/2026	Explained safe storage practices, including covered tanks and regular cleaning.	Recognized the need to protect stored rainwater from contamination.	
Friday 12/06/2026	Discussed groundwater recharge and the need to check site suitability before selecting a structure.	Learned that recharge depends on suitable soil, location, and water quality conditions.	
Saturday 13/06/2026	Summarized key messages and encouraged simple water-saving and maintenance practices.	Developed communication skills and identified practical conservation messages for households.	

WEEK 6: AWARENESS ABOUT WET AND DRY WASTE, RECYCLING, AND COMPOSTING

Objective of the Activity Done:

To demonstrate point-of-generation segregation and explain what happens after collection.

Detailed Work Description:

The team will use sample materials or clean demonstration items to show wet waste, dry recyclable waste,

sanitary waste, and domestic hazardous waste. A composting explanation will cover suitable organic

inputs, aeration, moisture balance, odour control, and the need to exclude plastics and hazardous items.

Recycling messages will stress keeping recoverable materials dry and using authorised channels.

Observations and Findings:

The team will collect feedback on which practices are easy, difficult, or unavailable. Recommendations

should consider bin space, collection schedules, willingness of residents, availability of composting space,

and local authorised recyclers.

Outcome and Learning:

Students learn to distinguish awareness from infrastructure, communicate health and safety points, and

build practical behaviour-change messages.

WEEK 6: AWARENESS ABOUT WET AND DRY WASTE, RECYCLING, AND COMPOSTING

DAILY ACTIVITY RECORD

DAY & DATE	BRIEF DESCRIPTION OF THE DAILY ACTIVITY	LEARNING OUTCOME	PERSON IN CHARGE / SIGNATURE
Monday 15/06/2026	Introduced the idea of separating waste at the place where it is generated.	Understood why segregation at source makes collection and recovery easier.	
Tuesday 16/06/2026	Used examples to explain wet waste such as food scraps, peels, and garden leaves.	Learned to identify biodegradable materials suitable for composting.	
Wednesday 17/06/2026	Explained dry recyclable materials such as paper, cardboard, plastic, glass, and metal.	Recognized recyclable materials and the importance of keeping them clean and dry.	
Thursday 18/06/2026	Discussed composting, including suitable organic inputs, moisture, and aeration.	Learned the basic conditions required for composting and odour control.	
Friday 19/06/2026	Explained separate handling of sanitary and domestic hazardous waste.	Improved awareness of safe handling and authorised disposal routes.	
Saturday 20/06/2026	Collected feedback and summarized practical segregation and recycling messages.	Learned to adapt awareness messages to available bins, collection schedules, and local facilities.	

WEEK 7: FOLLOW-UP OBSERVATIONS AND ASSESSMENT OF COMMUNITY AWARENESS

Objective of the Activity Done:

To revisit selected locations, assess whether key messages were understood, and identify immediate next

steps.

Detailed Work Description:

The team will repeat a short observation or feedback exercise at the same or comparable locations.

Follow-up questions may ask residents to identify waste categories, describe the purpose of a first-flush

unit, name one maintenance task, or explain where hazardous waste should go. The

responses with the baseline only if the same method was used.

Observations and Findings:

The report should present verified observations, not exaggerated impact claims. Useful evidence may

include attendance sheets, photographs of demonstrations, completed feedback forms, maintenance

checks, or community suggestions. Any change in behaviour must be labelled as reported, observed, or

measured.

Outcome and Learning: Students learn basic monitoring, comparison of baseline and follow-up information, and responsible

WEEK 7: FOLLOW-UP OBSERVATIONS AND ASSESSMENT OF COMMUNITY AWARENESS

DAILY ACTIVITY RECORD

DAY & DATE	BRIEF DESCRIPTION OF THE DAILY ACTIVITY	LEARNING OUTCOME	PERSON IN CHARGE / SIGNATURE
Monday 22/06/2026	Revisited selected locations and reviewed the main water and waste-management concerns discussed earlier.	Understood the purpose of follow-up observation and continuity in community interaction.	
Tuesday 23/06/2026	Asked residents to identify wet, dry, sanitary, and hazardous waste categories.	Reviewed awareness of common waste categories and segregation practices.	
Wednesday 24/06/2026	Discussed the purpose of first-flush units and filters in rainwater harvesting.	Checked understanding of basic rainwater harvesting components.	
Thursday 25/06/2026	Asked about routine maintenance tasks for gutters, filters, and storage tanks.	Recognized the importance of regular maintenance for safe and effective operation.	
Friday 26/06/2026	Recorded community feedback, suggestions, and any reported changes in practices.	Practised careful documentation and distinguished reported information from direct observation.	
Saturday 27/06/2026	Organized follow-up notes and compared them with the initial observations where appropriate.	Learned basic monitoring and responsible comparison of community feedback.	

WEEK 8: CONSOLIDATION, RECOMMENDATIONS, LEARNING OUTCOMES, AND FINAL REPORT
Objective of the Activity Done:

To consolidate documentation, identify realistic recommendations, reflect on learning, and prepare the

final report.

Detailed Work Description:

The team will organise daily logs, photographs, permission letters, survey forms, sketches, awareness

materials, and field notes. Findings will be grouped under water conservation, rainwater harvesting

feasibility, waste segregation, community awareness, and future action. The team will identify which

recommendations can be implemented immediately and which require local-authority or technical

approval.

Observations and Findings:

The final report will clearly separate background knowledge, actual observations, proposed actions, and

items requiring verification. All placeholders should be replaced or retained intentionally. Photographs

must include captions, dates, locations, and consent where required.

Outcome and Learning:

Students learn academic report preparation, evidence management, reflective practice, recommendation

writing, and the importance of honest documentation.

WEEK 8: CONSOLIDATION, RECOMMENDATIONS, LEARNING OUTCOMES, AND FINAL REPORT

DAILY ACTIVITY RECORD

DAY & DATE	BRIEF DESCRIPTION OF THE DAILY ACTIVITY	LEARNING OUTCOME	PERSON IN CHARGE / SIGNATURE
Monday 29/06/2026	Organized daily activity notes, survey records, photographs, and other project documents.	Improved evidence management and systematic project documentation.	
Tuesday 30/06/2026	Reviewed observations related to water conservation and rainwater harvesting feasibility.	Learned to separate general technical information from findings recorded in the community.	
Wednesday 01/07/2026	Reviewed information on household waste segregation, recycling, and composting.	Consolidated understanding of waste categories and practical management options.	
Thursday 02/07/2026	Identified recommendations that could be considered by households and community members.	Learned to prepare realistic recommendations based on the available observations.	
Friday 03/07/2026	Arranged the project activities, learning outcomes, and supporting records in report order.	Developed report-writing and presentation skills.	
Saturday 04/07/2026	Completed the final review of the report and summarized the learning from the project.	Understood the value of reflection, accurate documentation, and community participation.	

SURVEY QUESTIONS

1. Are you aware of water segregation and its importance?
Answer: Yes, water segregation helps separate different types of water for proper treatment and reuse.
2. Do you separate wastewater from rainwater at your home?
Answer: Yes, we try to keep rainwater separate from wastewater.
3. Does your house have a rainwater harvesting system?
Answer: No, our house does not currently have a rainwater harvesting system.
4. Do you know the benefits of rainwater harvesting?
Answer: Yes, rainwater harvesting helps conserve water and recharge groundwater.
5. How do you currently use rainwater at your home?
Answer: We use rainwater for gardening and cleaning whenever possible.
6. Do you face water shortages in your area?
Answer: Yes, we sometimes face water shortages, especially during summer.
7. Do you think rainwater harvesting can help reduce water shortages?
Answer: Yes, rainwater harvesting can help store water and reduce water shortages.

8. Is wastewater from your home reused for gardening or cleaning?
Answer: Yes, we sometimes reuse suitable household wastewater for gardening and cleaning.
9. Are you willing to install a rainwater harvesting system in your home?
Answer: Yes, I am willing to install a rainwater harvesting system to save water.
10. Would you participate in awareness programs about water segregation and rainwater harvesting?
Answer: Yes, I would participate in awareness programs to learn and share water conservation methods.

ENVIRONMENTAL AND COMMUNITY BENEFITS

- Improved understanding of seasonal rainfall, runoff, water conservation, and groundwater recharge.
- Potential reduction in avoidable runoff, local waterlogging, and erosion where suitable measures are implemented.
- Better recovery of recyclable materials and improved safety for sanitation workers through segregation.
- Reduced mixing of organic waste with dry and hazardous materials.
- Opportunities for household or community composting where space, management, and collection arrangements permit.
- Stronger community participation through shared responsibility, follow-up, and locally visible actions.
- Development of student skills in technical communication, field observation, documentation, teamwork, and ethical reporting.

Summary of Benefits

Environmental benefits	Community benefits
Reduces water pollution	Improves public health
Conserves freshwater	Improves water availability
Protects aquatic life	Reduces water expenses
Supports groundwater recharge	Maintains clean surroundings
Reduces flooding	Supports gardening and agriculture
Protects soil quality	Creates employment opportunities
Promotes sustainability	Encourages public awareness

CHALLENGES AND POSSIBLE SOLUTIONS

Challenge	Possible reason	Practical solution	Responsible stakeholder
Limited space for bins or tanks	Small plots / congested layout	Use compact labelled containers; identify shared locations	[Households / local body]
Mixed waste continues	Lack of awareness or irregular collection	Repeat demonstrations; align with collection schedule	[Residents / collector]
Filter or gutter blockage	Leaves, dust, lack of inspection	Pre-monsoon and periodic cleaning checklist	[Owner / institution]
Fear of contaminated harvested water	Unclear end use or maintenance	Promote suitable non-potable use unless treated and tested	[Project team / technical adviser]
No authorised route for hazardous waste	Information gap	Coordinate with local authority / authorised take-back channel	[Local body / residents]
Low participation	Timing, trust, competing priorities	Use local volunteers and shorter repeated sessions	[Students / community leaders]

LEARNING OUTCOMES

- Explain the purpose and working sequence of a basic rooftop rainwater harvesting system.
- Differentiate storage, filtration, surface-runoff management, and groundwater recharge.

- Identify the importance of first flush, overflow, maintenance, and safe end use.
- Classify common household waste into wet, dry recyclable, sanitary, and hazardous streams.
- Describe basic composting and recycling practices without claiming that every material is recyclable everywhere.
- Conduct respectful community interaction and document observations without fabricating results.
- Prepare tables, activity logs, photographs, and recommendations in a formal academic report.
- Reflect on teamwork, communication, technical understanding, and community responsibility.

Expected Outcomes

- Better understanding of different types of water.
- Prevention of unnecessary mixing of clean water and wastewater.
- Improved opportunities for wastewater treatment and reuse.
- Reduced freshwater wastage.
- Increased awareness of water conservation and environmental protection.

CONCLUSION

Rainwater harvesting and waste segregation are practical, complementary approaches to community sustainability. Rainwater harvesting addresses the way rainfall is captured,

stored, filtered, or returned to the ground, while waste segregation addresses the way materials are handled after use. Together, they reduce avoidable resource loss and help protect drains, soil, groundwater, and public spaces.

The success of the project depends on local suitability, clear responsibilities, routine maintenance, reliable collection systems, and continued community participation. The student team should therefore present recommendations that are technically appropriate, affordable where possible, and supported by actual evidence. The final version of this report must replace all placeholders with verified project details and retain a transparent distinction between observed results, community-reported information, and proposed actions.

For communities, it improves sanitation, helps managing, water shortages ,reduces water expenses, and encourages public participation in environmental protection. Therefore, implementing proper water segregation systems in homes, schools,industries, and public places is an important step towards a cleaner environment, healthier communities, and sustainable water management..

Student Self Evaluation for the Community Service Project

Student Name : Tadepalli Uma Naga Vyshnavi

Roll Number : 24471A05CV

Term of CSP : MAY2026 TO JULY2026

Date of Evaluation :

Name of the Person in-charge : Y.Chandana

Address with Mobile Number : Narasaraopeta,BuremPeta 9959061400

Please rate Student's performance in the following areas:

Please note that your evaluation shall be done independent of the student's self-evaluation.

1	Oral Communication	1	2	3	4	5
2	Written Communication	1	2	3	4	5
3	Proactiveness	1	2	3	4	5
4	Interaction Ability	1	2	3	4	5
5	Positive Attitude	1	2	3	4	5
6	Professionalism	1	2	3	4	5
7	WorkPlan and Organisation	1	2	3	4	5
8	Self Confidence	1	2	3	4	5
9	Overall Performance	1	2	3	4	5

Date: Signature of Student

Evaluation by the Person in-charge in the Community Habitation

Student Name : Tadepalli Uma Naga Vyshnavi

Roll Number :24471A05CV

Term of CSP : MAY2026 TO JULY2026

Date of Evaluation :

IN-CHARGE NAME :Y.CHANDANA

Address&Number :Narasaraopet,Burempet,9959061400

Please rate Student's Performance in the following areas

1	Oral Communication	1	2	3	4	5
2	Written Communication	1	2	3	4	5
3	Proactiveness	1	2	3	4	5
4	Interaction Ability	1	2	3	4	5
5	Positive Attitude	1	2	3	4	5
6	Professionalism	1	2	3	4	5
7	WorkPlan and Organisation	1	2	3	4	5
8	Self Confidence	1	2	3	4	5
9	Overall Performance	1	2	3	4	5

Date: Signature of Supervisor

EVIDENCE


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Water Segregation and Rainwater Harvesting
Awareness & Conservation Portal

STUDENT DEMO PORTAL

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WATER SEGREGATION & RAINWATER HARVESTING

Water Segregation & Rainwater Harvesting

Fresh water is finite, yet everyday habits lose millions of liters of clean runoff and greywater. This academic community project provides actionable blueprints for source water segregation, low-cost rooftop rainwater harvesting, and grassroots conservation habits.

Learn About Water Segregation | Rainwater Harvesting | Water Saving Tips



Catch Rain • Segregate Drains • Conserve Life
Decentralized water stewardship for residential communities

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PILLAR 1 • WASTEWATER SEGREGATION

What is Water Segregation?

Water segregation is the practice of separating domestic wastewater at the point of generation based on contamination levels, preventing relatively clean wash water from mixing with pathogen-heavy sewage.

The Problem with Single-Pipe Drainage

When bathing or laundry water is combined into a single sewer pipe with toilet waste, 100% of the household output is immediately contaminated and categorized as hazardous blackwater. Segregating greywater at the source enables up to 70% of wastewater to be cleaned with simple natural filters and used for gardening or toilet flush recharge.

The 3 Domestic Wastewater Streams

1. Greywater (Light)

Comes from bathroom washbasins, showers, bathtubs, and washing machine rinse cycles. Contains minor soap, shampoo, lint, and hair.

Pathogen Level: Low
Ease of Reuse: Very High (Garden, flushing)

2. Blackwater (Heavy)

Comes from flush toilets, urinals, and food preparation sinks with heavy grease and raw animal residues. High in feces, harmful bacteria, and nitrogen.

Pathogen Level: Extremely High (E. coli)
Treatment: Septic tank or biological STP

3. Stormwater Runoff

Rainwater that runs across driveways, paved pathways, and rooftops during rainfall. May contain silt, leaves, and airborne particulates.

Pathogen Level: Negligible
Treatment: Sediment filter & percolation pit

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PILLAR 2 • ROOFTOP RAINWATER HARVESTING

Rainwater Harvesting (RWH)

Rainwater harvesting is the technique of capturing, filtering, and storing precipitation from rooftops for direct consumption, domestic non-potable chores, or recharging declining groundwater aquifers.

INTERACTIVE SYSTEM DIAGRAM

Roof → Pipe → Filter → Storage Tank → Reuse

Click on any step below to explore its role, setup requirements, and maintenance checklist:

1. Roof — Catchment
2. Pipe — Conveyance
3. Filter — First Flush & Mesh
4. Storage Tank — Cistern / Pit
5. Reuse — Garden & Chores

STAGE 1: THE ROOF (CATCHMENT SURFACE)

How Rooftop Collection Works

The roof is the initial surface directly receiving monsoon rainfall. Smooth concrete RCC roofs, clay tiles, and corrugated GI sheets provide optimal runoff efficiency (80% to 90%). Keep rooftop clean of bird droppings, dry leaves, and moss prior to the rainy season.

PILLAR 3 • COMMUNITY ACTION

Water Saving Tips & Habits

Conservation doesn't require high-tech gear. Small daily habit changes across households accumulate to save millions of liters of clean water annually.

1 Turn off taps when not in use

Always turn off the faucet while brushing teeth, shaving, soaping hands, or washing utensils. A tap running freely expels 6 to 8 liters every minute.

Saves ~12 L per brush

2 Repair leaking taps and pipes promptly

A tiny leak of one drop per second squanders over 30 liters daily (10,000+ liters annually). Inspect washers, pipe joints, and toilet flush valves at least once a month.

Saves 30–50 L / day

3 Reuse water whenever possible

Do not discard the rejected discharge line from domestic RO water purifiers; route it into a bucket to mop floors or clean balconies. Use rinse water from vegetables to water houseplants.

Saves 40–80 L / day

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PROJECT PROFILE

About Water Segregation and Rainwater Harvesting

This project was created to spread community awareness about water conservation, wastewater segregation at source, and rooftop rainwater harvesting.

Project Scope & Community Motivation

As water scarcity escalates in both rural and urban areas, groundwater depletion has reached critical levels. Traditional centralized water infrastructure often fails during hot summers. Our goal is to show local residents and students that sustainable water management begins at home with decentralized, affordable solutions:

Objective 1

Educate families on segregating greywater from blackwater to recycle wash water for home gardens.

Objective 2

Demystify rooftop rainwater harvesting with intuitive diagrammatic flows and volume calculators.

Objective 3

Provide practical, low-cost daily habits that prevent dripping faucet waste and careless usage.